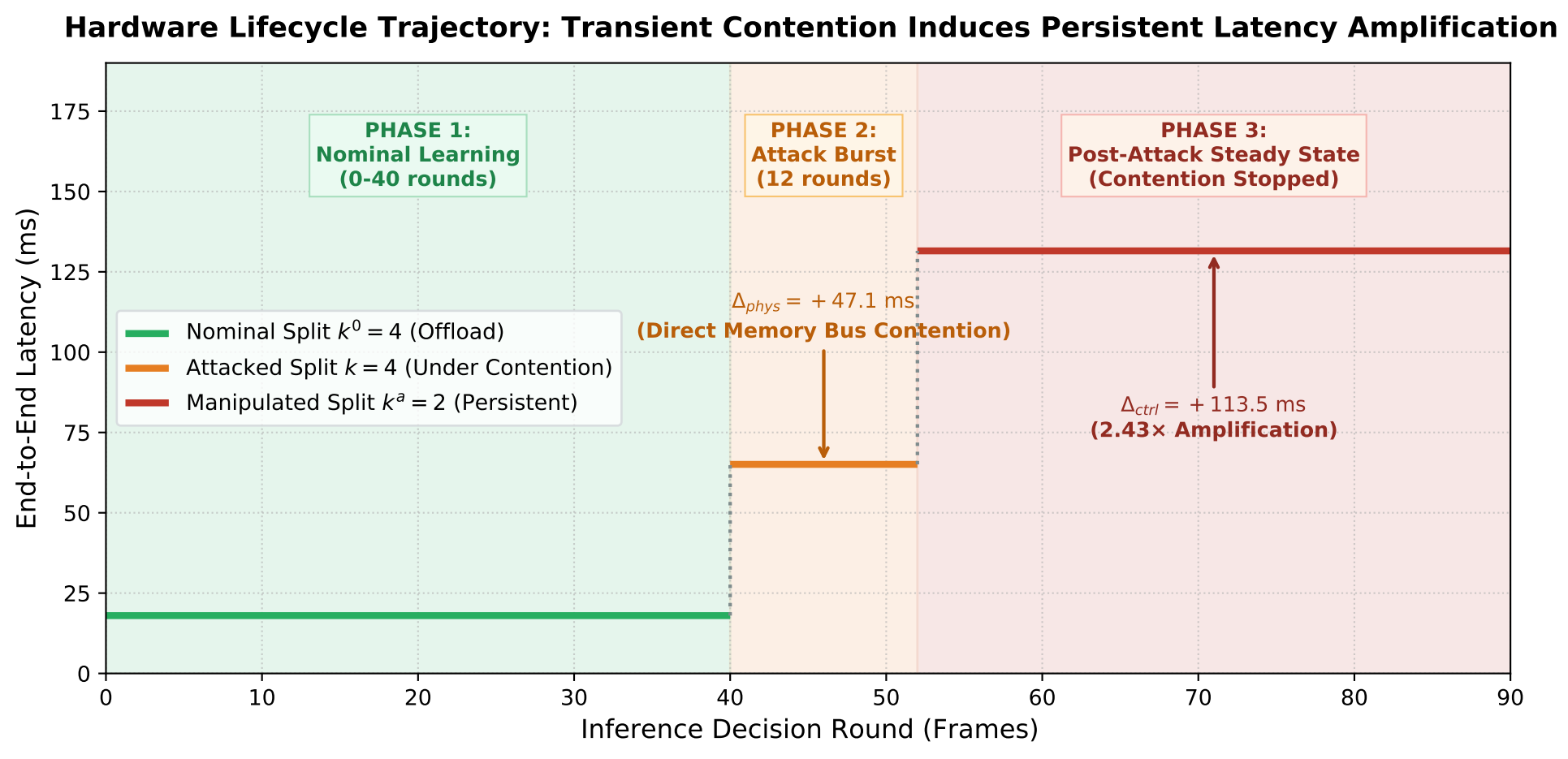
Hardware Lifecycle Trajectory: Transient Contention Induces Persistent Latency Amplification
PHASE 1:
Nominal Learning
(0-40 rounds)
PHASE 2:
Attack Burst
(12 rounds)
PHASE 3:
Post-Attack Steady State
(Contention Stopped)
Nominal Split $k^0 = 4$ (Offload)
Attacked Split $k = 4$ (Under Contention)
Manipulated Split $k^a = 2$ (Persistent)
$\Delta_{phys} = +47.1$ ms
(Direct Memory Bus Contention)
$\Delta_{ctrl} = +113.5$ ms
(2.43× Amplification)
End-to-End Latency (ms)
Inference Decision Round (Frames)
0
25
50
75
100
125
150
175
0
10
20
30
40
50
60
70
80
90